Exploring Decentralized Prediction Markets: Accuracy, Skill, and Bias on Polymarket

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Abstract

Prediction markets aggregate dispersed information into probabilistic forecasts that can guide decision-making and risk management. The recent emergence of decentralized, blockchain-based platforms thus gives practitioners access to real-time probabilities for a wide range of events and, at the same time, generates unprecedented amounts of real-world trading data that are highly valuable for economic research. In this study, we introduce this novel data source and explore several foundational questions regarding accuracy, skill, and bias in decentralized prediction markets using a dataset of more than 124 million trades from the leading platform Polymarket. At the market level, we document a tendency to overtrade the default and "Yes" option but find no evidence of a general longshot bias. Overall, market prices closely track realized probabilities and slightly outperform bookmaker odds. Inaccuracies primarily occur early in a contract's lifecycle and close to resolution, indicating that markets require time to incorporate new information. At the individual level, only 30% of traders earn positive profits, and this share decreases over time. We also provide evidence of skilled traders: the proportion of traders with positive gains per bet exceeds expectations under randomness, and traders' profits are persistent over time. Lastly, we study differences in characteristics between successful and unsuccessful traders and show how they contribute to trading performance. Our results suggest that skilled traders may generate profits by exploiting the biases of less skilled market participants. We discuss implications for managers, market participants, and regulators, and outline promising areas for future research across several disciplines.

Keywords: prediction markets; behavioral biases; forecasting accuracy; event markets; market efficiency

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1 Introduction

Economists have long hailed prediction markets as a promising tool for improving decisions and, consequently, social welfare in many domains (Arrow et al., 2008). Because market participants are financially incentivized to aggregate and discover information to accurately predict events, prices on prediction markets are often considered highly accurate and have been shown to outperform opinion polls (Berg et al., 2008; Wolfers and Zitzewitz, 2004). Thus, public prediction markets that efficiently process all available information on important future events, such as elections, geopolitical risks, or macroeconomic developments, could be very useful to decision-makers for risk management and resource allocation (Berg and Rietz, 2003). However, until recently, prediction markets remained a niche phenomenon, unable to fulfill these high expectations due to a lack of technical capabilities and regulatory hurdles.

This has changed in the past five years with the emergence of a new generation of prediction markets that target a mass audience and offer much higher liquidity such as Kalshi and Polymarket. Furthermore, financial institutions are also increasingly entering the event-contract space. For example, the trading app Robinhood began offering customers the ability to trade event contracts since the U.S. presidential election in October 2024 (Robinhood, 2024). This offering was significantly expanded in 2025 through a partnership with Kalshi, and users traded more than 2 billion contracts until August 2025 (Robinhood, 2025). Similarly, Intercontinental Exchange announced a strategic partnership with Polymarket in October 2025, including an investment commitment of up to USD 2 billion and plans to distribute Polymarket's event-driven data across its institutional customers (ICE, 2025).

Beyond their promise for improving decision-making, the transparency of these new, decentralized prediction markets provides researchers with a unique window into real-world financial behavior and biases. In contrast to traditional prediction markets or financial markets, every trade is publicly documented on the blockchain. The unprecedented scale and granularity of these trading data open up vast new possibilities for economic research in the coming years. In this study, we use a dataset from the leading platform Polymarket that contains more than 124 million trades by nearly one million traders with a total volume of USD 48 billion. Drawing on this novel data source, our study offers a first systematic overview of decentralized prediction markets and explores a range of timely research questions along two overarching dimensions.

First, at the market level, we analyze the accuracy of decentralized prediction markets by assessing the extent to which market prices reflect true event probabilities and by comparing their predictive performance to that of traditional bookmakers. We show that, although certain biases exist, such as a tendency to overtrade the default and "Yes" option, market predictions are accurate most of the time. Notably, we do not find evidence of a general longshot bias. Deviations from accuracy primarily arise in early trades and trades executed close to event resolution, which indicates that markets require time to process and incorporate new information. Furthermore, Polymarket's tennis predictions are slightly better than corresponding bookmaker odds, suggesting that decentralized prediction markets can generate highly accurate estimates.

Second, the fact that trades are linked to wallet addresses enables us to study decision-making at the individual level. Such analysis is usually only possible with experimental data, which generally involves lower stakes and far fewer participants than the close to one million traders in our sample. Therefore, our data introduces a new empirical dimension to research on betting behavior and provides a richer understanding of market participants and their role in how prediction markets function. We show how profits and losses are distributed, and that only about 30% of traders earn profits, with this share decreasing over time. Furthermore, we provide evidence that skilled traders exist. The proportion of traders with positive gains per bet exceeds what would be expected under pure randomness, and profits and losses are persistent over time. Moreover, our analyses of the differences between top and bottom traders indicate that skilled traders may exploit biases of unskilled traders, which contributes to their success.

Overall, our study contributes to several research areas. In behavioral economics, we investigate the decision-making of nearly one million traders. In financial markets, we provide a detailed analysis of how decentralized prediction markets operate. Our findings also have implications for risk management, as managers can use market predictions to assess risks and bets to hedge them. For platform economics, our results inform discussions of market design, regulation, and consumer protection. Beyond these contributions, our study opens and presents promising research avenues across several disciplines.

The remainder of the study is structured as follows. Section 2 defines and explains decentralized prediction markets and reviews related literature. Section 3 describes the dataset. As discussed above, the results of our analysis are presented in two parts: Section 4 examines accuracy on the market level, while Section 5 focuses on individual trading behavior. Section 6 discusses the implications of our findings, and promising research avenues. Section 7 concludes.

2 Theory and literature

2.1 Defining (decentralized) prediction markets

Prediction markets are markets in which participants can trade contracts whose payoffs are based on the outcomes of uncertain future events (Arrow et al., 2008). Most prediction markets use "winner-take-all" contracts, which means that the contracts pay out a predetermined amount if the event in question occurs (Wolfers and Zitzewitz, 2004). For example, if traders want to bet on a specific candidate winning an election, they can buy a contract for $p that pays $1 if the candidate wins (event Y). Thus, the market price p indicates the market's expectation of the probability p(Y) that event Y will occur.

For prediction markets to function properly, the trust of market participants in the reliability of the market is essential. Thus, the conditions under which an event is considered to have occurred must be clearly defined, and payment for a winning prediction must be guaranteed. Then, monetary incentives should motivate participants to use all available information to gain an informational advantage and profit from mispriced probabilities. From a theoretical point of view, therefore, prices on an efficient prediction market should be the best predictor of future events because they reflect the expected probabilities based on all available information at that time (Wolfers and Zitzewitz, 2004). If insider trading is possible, prediction markets would also incorporate private information, since insiders have strong incentives to exploit their informational advantage, which corresponds to Fama's (1970) strong form of market efficiency.[3] It is not necessary for every market participant to be rational and free of behavioral biases for (nearly) efficient prediction markets; rather, it only requires that prices are set at the margin by sufficiently many profit-oriented and informed traders to eliminate inefficiencies (Grossman and Stiglitz, 1980; Forsythe et al. 1992).

In decentralized prediction markets, the reliability of the settlement process is not guaranteed by a central operator but is implemented via smart contracts on a public blockchain. This means that the rules governing trading, settlement and payout are encoded and automatically enforced, which reduces counterparty risk and ensures transparency. In addition, platforms such as Polymarket use stablecoins that are pegged to the U.S. dollar, so that a "winner-take-all" share represents a claim to USDC 1 if the event occurs and 0 otherwise (Polymarket, 2025). This preserves the usual interpretation of prices as probabilities, while traders are not affected by the high short-term price volatility of unpegged cryptocurrencies.

Decentralized prediction markets must also solve the problem of clearly determining whether an event has occurred without relying on a central authority (e.g., bookmakers). In the case of Polymarket, each market is accompanied by public "market rules" specifying the relevant event and data sources, and resolution is handled by a decentralized oracle system (Cong et al. 2025). Users need to post a bond of $750 to propose an outcome that is automatically accepted as the final outcome for all contracts in the respective markets if it is not challenged within two hours. If challenged by another user with another $750 bond, the oracle escalates the decision, and token holders vote on the issue after a debate period. Since repayment of the bond depends on proposing the correct outcome, this system incentivizes accurate reporting. Once the outcome is finalized, the smart contract automatically settles the market by allowing holders of winning shares to redeem them for USDC 1, while losing shares expire worthless (Polymarket, 2025).

[3] A recent example of suspected insider trading on Polymarket occurred when the price of contracts on María Corina Machado winning the 2025 Nobel Peace Prize jumped from 3.6% to 73% just hours before the Nobel Committee announced her victory on October 10, 2025 (Kochkodin, 2025).

Trading prior to resolution typically occurs via an order book in which limit orders to buy or sell outcomes determine the current market price. Due to the binary nature of markets on Polymarket, the order book also displays the mirrored order for the opposite outcome. For example, if trader A submits a buy order for a "Yes" share at \$0.30 (below the current market price), it will appear as a bid at \$0.30 in the "Yes" order book and as an ask at \$0.70 in the "No" order book. This inversion of binary outcomes implies three possible ways of matching orders. First, another user (trader B) can sell a "Yes" share at \$0.30, resulting in a transfer of the purchase price and the share between the traders. Second, trader B can buy a "No" share at \$0.70, in which case new shares are minted and distributed to both traders (i.e., A pays \$0.30 and receives a "Yes" share, and B pays \$0.70 and receives a "No" share). Third, if both traders want to sell opposite positions, the shares are merged and burned, freeing \$1 in collateral (i.e., A receives \$0.30 for selling "Yes" and B receives \$0.70 for selling "No"). Partial executions and combinations of these three variants are also possible, and traders can place market orders that are executed immediately against the existing order book. Furthermore, until November 2025, Polymarket did not charge any direct trading fees or impose trading size limits (Polymarket, 2025). This combination of smart contracts based on stablecoins, decentralized outcome determination via oracles, and publicly traceable transactions allows decentralized prediction markets to function without a central authority, even though in practice such platforms may still rely on some centralized components (e.g., web interfaces).

Table 1 summarizes the key characteristics of decentralized prediction markets and compares them to other common ways of betting on future events. The table highlights that the transparency of trading data and the absence of central counterparties like bookmakers or clearing houses are a main difference. Because extensive trader-level data were rarely available outside laboratory settings, previous empirical literature on trader and bettor behavior has focused primarily on market-level outcomes, for example in prediction and bookmaker markets (Spann and Skiera, 2009), parimutuel betting (Weitzman, 1965), and options markets (Gurevich et al., 2009). A further difference is that prediction markets can, in principle, be used to bet on almost any event, whereas bookmakers primarily concentrate on sporting events and derivatives

	Decentralized prediction markets	Betting exchanges and centralized prediction markets	Betting with bookmakers	Financial derivatives (options and futures)
Example	Polymarket	Betfair, Smarkets (exchanges); PredictIt, Kalshi (prediction markets)	Bet365, DraftKings, William Hill	Equity and commodity options or futures
Pricing mechanism	User-set via order books or automated market makers	User-set via order books	Bookmaker sets the odds	Market-determined option premiums
Counterparty	Other users via smart contracts	Other users; platform intermediates	The bookmaker ("the house")	Clearing house; other traders or market makers
Custody and collateral	Crypto collateral in smart contracts	Fiat (sometimes crypto) custodied by venue	Fiat staked with bookmaker	Cash/margin at broker; standardized collateral rules
Settlement	On-chain oracle reports event result	Venue settles per house rules	Bookmaker settles per house rules	Payoff at expiration or exercise based on underlying price
Typical underlying	Real-world events (politics, sports, prices of financial assets, …)	Mostly sports (exchanges) and events (prediction markets)	Mostly sports	Financial assets
Payout profile	Binary ($1 if correct, $0 if not)	Fixed-odds (exchanges) or binary event contracts (prediction markets)	Fixed-odds	Nonlinear, asymmetric payoffs (options)
Cost	Platform fees and network gas	Commissions on net winnings or trading fees	Included in odds (overround)	Broker commissions, exchange fees
Transparency	Prices and order book visible; all trades visible on-chain (pseudonymous)	Odds and order book visible; platform controls trading data	Public odds; opaque margins	Standardized contracts and public prices
Secondary trading	Yes	Yes	Typically no	Yes
Price interpretation	Price ≈ implied probability	1/odds ≈ implied probability (exchanges); prices ≈ probabilities (prediction markets)	1/odds ≈ implied probability	Premiums imply risk-neutral probabilities via option pricing models

Table 1: Comparison of different ways of betting on future events. For broader comparisons of betting and forecasting methods, see, for example, Spann and Skiera (2009) or Franck et al. (2010).

require a tradable underlying asset with an observable market price. Finally, while prediction markets contract prices can be interpreted directly as probabilities of future events, translating bookmaker odds with overround (see Section 4.2) or option prices into probabilities is more complex (e.g., Black and Scholes, 1973).

2.2 Related literature

As discussed before, prediction markets have a long history in the economic literature due to their potential to efficiently aggregate information (Hayek, 1945; Surowiecki, 2004; Arrow et al., 2008). Thus, a large share of empirical research on traditional prediction markets has focused on the accuracy of the predictions compared to other information sources such as bookmakers, tipsters, experts or polls (e.g., Franck et al., 2010; Atanasov et al. 2017). Most of these studies are concerned with markets predicting political events, and in particular U.S. elections (Forsythe et al. 1992; Snowberg et al., 2007; Berg et al. 2008; Goodell et al. 2020). Another research stream discusses the potential for corporate prediction markets as decision support systems (Berg and Rietz, 2003; van Bruggen et al. 2010; Cowgill and Zitzewitz, 2015). Furthermore, some studies examine how the accuracy of prediction markets can be affected by behavioral biases or outcome manipulation (Ottaviani and Sørensen, 2007; Healy et al., 2010; Rothschild, 2009; Page and Clemen, 2013).

Due to the novelty of the topic, the literature on modern, decentralized prediction markets is still in its infancy and consists almost exclusively of working and conference papers. Most of these focus on market accuracy as well, for example in the 2024 U.S. presidential election (Cutting et al., 2025; Sethi et al., 2025) or in measuring the credit risk of government bonds (McGurk and Becker, 2025). However, some studies have begun to utilize trader-level data, such as Eichengreen et al. (2025), who use trading behavior on Polymarket to distinguish between hawks and doves in the context of monetary policy decisions. Others examine arbitrage opportunities arising from mispricing in separate but correlated markets on Polymarket (Saguillo et al., 2025) or between different prediction market platforms (Ng et al., 2025).

These recent studies demonstrate the potential of decentralized prediction markets both for economists, who can learn about betting and trading behavior, and for practitioners, who can use them as a data source for decision-making. However, most existing work focuses on only a small number of events and is concerned with specialized research questions. Our study takes a broader perspective by using an extensive dataset of more than 124 million trades from 68,000 different markets on Polymarket. Using this dataset, we provide a first overview of both the overall accuracy of prices on decentralized prediction markets and betting and trading behavior at the individual level. In doing so, our study paves the way for future, in-depth research in this novel field.

3 Data and methods

3.1 Data Sources

A key feature of decentralized prediction markets is the transparency of on-chain activity, since settlements and transfers are publicly visible on the blockchain. Additionally, Polymarket provides traders and researchers with various application programming interfaces (APIs) for more convenient access to large amounts of trading data. To obtain a dataset of all activities on Polymarket, we use a three-step approach: First, we use the Polymarket Gamma API to get a list of all markets and their outcomes until September 2025. Second, we utilize the trades endpoint of the Polymarket Data API to obtain a list of trades on these markets. As only the last 10,500 trades can be accessed for each market via the API, we use the ids of the wallets involved in these trades to get the complete history of each user via the activity endpoint. These activities can include the merging or splitting of tokens, liquidity rewards, yields for long-term bets, or payments after winning the bet. We focus primarily on trades, as we are interested in market prices and profits of traders due to betting or trading without external incentives such as rewards. Our final dataset contains more than 124.5 million trades made by approximately 974,000 different traders between November 2022 and September 2025. The total volume of the bets is USDC 48 billion distributed across 68,000 different predictions of events, called markets.[4] Figure 1 visualizes the monthly development of Polymarket based on our dataset. It depicts a significant increase in activity during 2024, and particularly the presidential election campaign. In 2025, the number of monthly active markets continued to grow, though trading activity decreased.

[4] Our dataset should be almost complete for the analyzed period. Due to the limitation of the trades endpoint which we used to identify the active users on Polymarket, it is possible that we have missed the activities of some small and infrequent traders. However, this likely only affects a small minority of traders who exclusively bet early in highly active markets. Another indicator of the near completeness of our dataset is that our trading volume is 1.86 times greater than the reported volume of all markets created and closed during this period. Since we consider both sides of each trade, our dataset contains approximately 93% of the total activity. Thus, the probability that a trade is completely missing for both sides should be significantly lower than 7%.

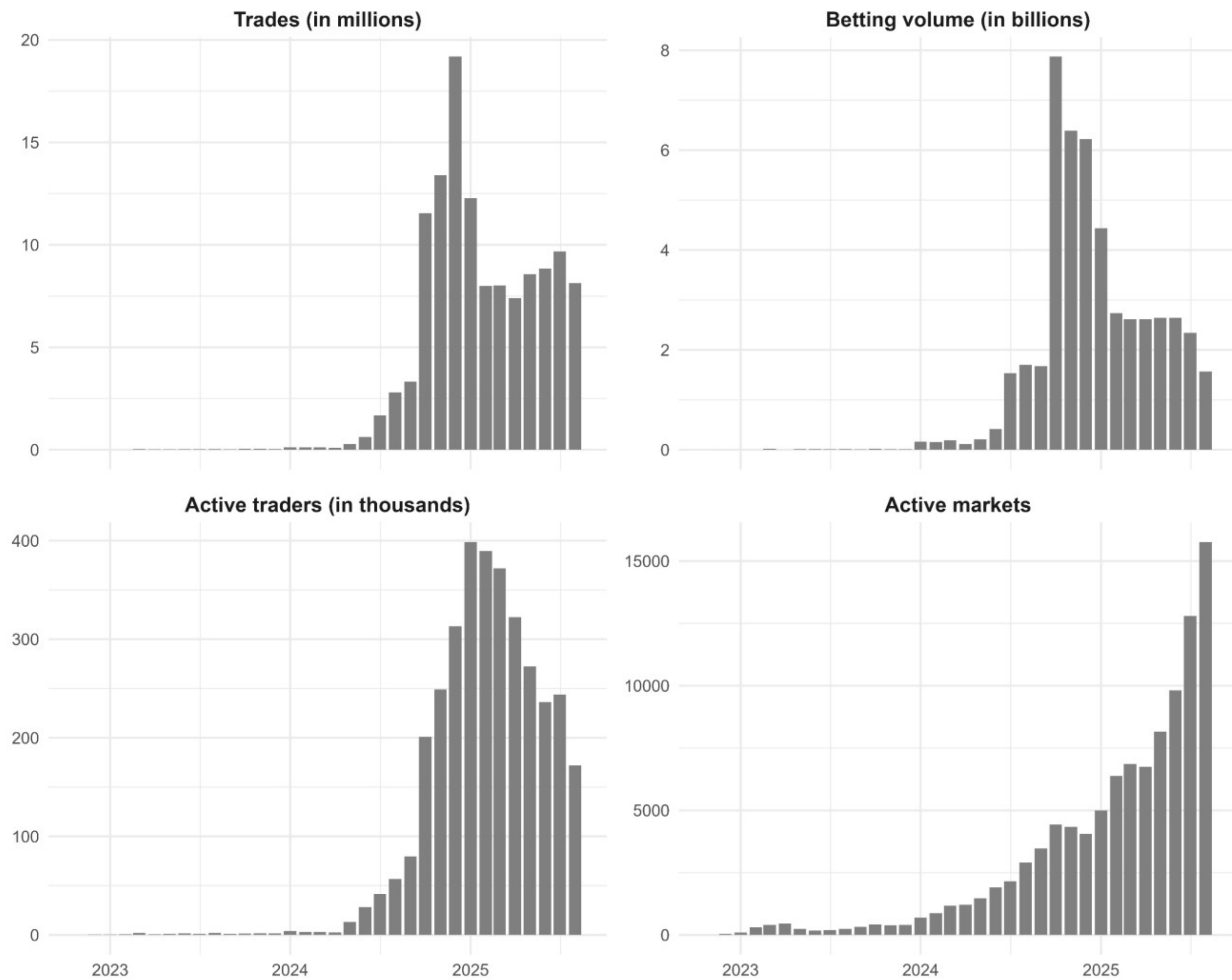

Figure 1: Overview of the monthly development of Polymarket according to our dataset. There is an increase in activity during 2024 (U.S. presidential election). In 2025, the number of monthly active markets continues to increase but trading activity decreases.

Each trade in our dataset features a timestamp, a market id and question representing the bet, such as “Will player A win their match against player B on date D?”, an outcome, such as “Yes” and “No”, and the side of the trade, i.e., “BUY” or “SELL”. The price reflects both the amount exchanged between traders and the market’s implied probability of the event. For example, a price of USDC 0.80 on a given market for a bought “Yes” share implies a probability of 80% that player A will win. Note that the payout when winning a bet is USDC 1, and 0 otherwise. Bet size is stored in the variable *size* that indicates the number of contracts. For instance, for the given bet and price, buying 100 contracts would mean betting USDC 80 on player A winning, with a payout of USDC 100 if A wins and a payout of 0 if A loses. Therefore, the total profit when A wins is USDC (1 – 0.80) x 100 = USDC 20, while the possible loss is USDC 80. Selling this prediction is like betting that A will not win: If A does not win the profit is USDC 80, and if A wins, the loss is USDC 20 (see also Section 2.1 for a discussion of the binary nature of the order book).

Finally, we need to determine whether the predictions have been correct, i.e., whether the bets have been successful. For this purpose, we use the variable "outcomePrice" from the list of markets discussed above. It indicates whether the price at settlement is USDC 1 (prediction has been correct), or 0 (prediction has been wrong). To ensure the validity of this variable, we manually check and correct all outcome prices that are not 0, 0.5 or 1 when rounding to two decimal places.[5] A payment of USDC 0.5 for both outcomes appears only in rare cases when neither of the possible outcomes occurred (e.g., a tennis game is canceled).

3.2 *Computing profits and losses for traders over time*

As briefly discussed in the previous subsection, the profit (or loss) of a trade is the difference between the price paid for the contract and the payment when redeeming it (usually 0 or USDC 1). For example, if a trader purchases a "No" share for USDC 0.30, the potential profits are USDC 0.70 or - USDC 0.30, depending on the outcome of the event. Note that, due to the binary nature of Polymarket's order book, selling an outcome is equivalent to buying the complementary outcome.

Thus, we only need to sum up the profits of all trades on closed markets to calculate the total profits and losses ("PnL") per trader until a certain date. For open markets, we must value the contracts according to current market prices. As we do not have access to the current order book at each point in time, we use the price of the last trade in the respective market as a representation of the current market price. Since it is possible that one outcome is more frequently traded than the other, we use the binary price logic to convert the last trade in the market to the respective outcome.[6]

[5] Specifically, this affects some soccer and NFL games from February 2023, which are likely database errors. In a few cases, the total payouts amounted to two, due to none of the outcomes occurring. Since each contract received half of the collateral, we adjusted the value to 0.5. Therefore, the total payouts amount to USDC 1 in all cases. In addition, we also checked whether the results are consistent with the payouts from redeemed contracts sourced from the activity endpoint.

[6] We cross-checked our PnL calculations with data from Polymarket's website and Polymarket Analytics (polymarketanalytics.com) across a sample of 100 random, top, and bottom traders. Results were largely consistent,

As August 2025 is the last month with complete trading data in our sample, we use September 1, 2025, as the last date to calculate the total PnL of all traders. We then repeat this procedure for each month with available trading data for all traders. This allows us to analyze the development of profits over time and to compute monthly profits, which are the changes in total PnL compared to the previous month.

3.3 Computing trader characteristics

To investigate the potential relationship between trader characteristics and the achieved profits and losses in Section 5.4, we also compute summary statistics for each trader. Specifically, we calculate each trader's fraction of trades within certain probability ranges (e.g., higher than 99%, or less than 25%), and certain timing percentiles (e.g., the first x% of market duration). Note that we account for sales by using (1 – price) as a representation of the implied probability, as selling a prediction implies betting on the other outcome. We calculate these fractions with weights based on the number of contracts per bet. Additionally, we compute the contract-weighted mean probability and variance, the fraction of trades within quantiles of market duration (i.e., the trade takes place in the first x% of the duration of this market), the fractions of bets on the outcome "Yes" (for bets with these outcomes), and the default outcome, respectively, the total number of trades, markets, traded contracts, and traded USDC, and the USDC amount per bet.

4 Market level: Accuracy and market-wide bias

4.1 Are market prices a good estimate for the probability of future events?

We begin our analysis by examining whether prices, on average, provide good estimates of the true real-world probabilities of events. In particular, we investigate whether these prices are affected by systematic distortions in probability assessments, such as the favorite-longshot bias (Ali, 1977). To study the evolution of accuracy over time, we observe each market at ten points that are evenly spaced between the first and

with small differences attributable to minor methodological differences, rewards, or pre-November-2022 trades excluded from our data. These checks give us confidence that our methodology accurately reflects trading performance on Polymarket.

the last trade. At each point, the market price is calculated as the mean price of the last ten trades. We also differentiate between trades where traders bet on the default option on Polymarket (often "Yes") versus trades on the other option.

To estimate the unknown true probabilities, we follow the common approach in the betting literature (e.g., Snowberg and Wolfers, 2010; Krčál et al., 2016) by grouping observations and measuring the relative frequency of the winning outcome. Specifically, for each timing decile and outcome type, we sort observations by their market price and split them into 50 equally sized groups. We then compare the average market price within each group to the corresponding empirical probability of the winning outcome. Figure 2 reports the results for all ten observation points.

Take, for example, the first subplot, which shows the difference between the mean implied probability and the observed probability for all price groups after 10% of the trading time has elapsed. The subplot indicates that shortly after the creation of a new market, prices for betting on the default option are too high (dark grey), while prices for the complementary outcome are too low (light grey). This effect is particularly pronounced for low implied probabilities of the complementary outcome (price too low) and high implied probabilities of the default outcome (price too high). In addition, the likelihood of events with very low probabilities is underestimated. Over time, these distortions decrease, and we no longer observe systematic mispricing across probability levels.

At the end of the trading period, however, an S-shaped pattern emerges that could be consistent with a very strong longshot bias or probability weighting as in prospect theory (Kahneman and Tversky, 1979). A more plausible explanation in our setting is that new information quickly becomes available near the end of the market (see also the next subsection). This could be, for example, the victory of a candidate in an election or a team in a sporting event. Since some traders naturally obtain and process this information faster than others, they can use the existing order book (in which the prices of limit orders are too low for the winning outcome and too high for the losing outcome) to place bets at favorable prices, generating the observed pattern.

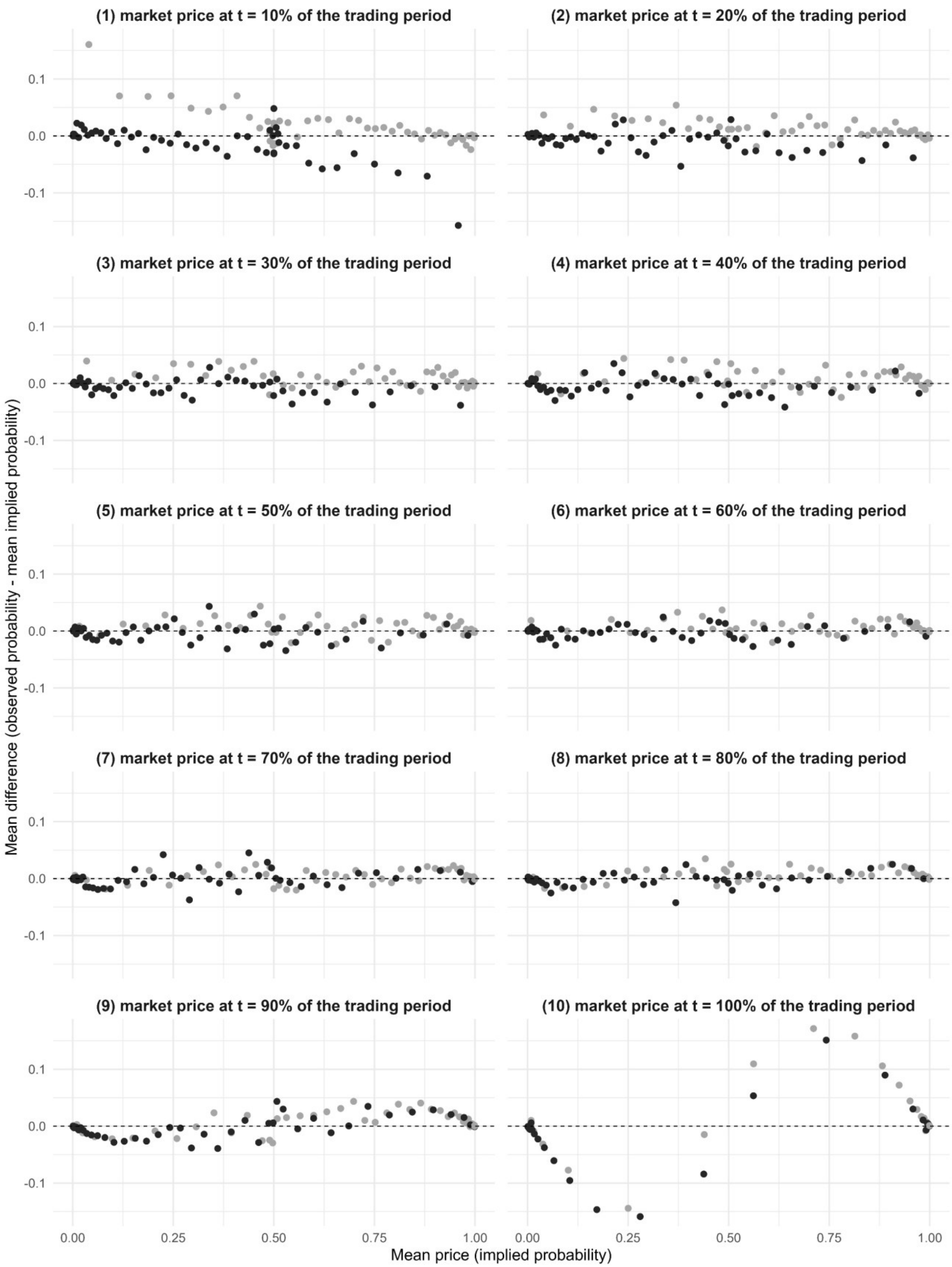
(1) market price at t = 10% of the trading period
(2) market price at t = 20% of the trading period
(3) market price at t = 30% of the trading period
(4) market price at t = 40% of the trading period
(5) market price at t = 50% of the trading period
(6) market price at t = 60% of the trading period
(7) market price at t = 70% of the trading period
(8) market price at t = 80% of the trading period
(9) market price at t = 90% of the trading period
(10) market price at t = 100% of the trading period
Mean difference (observed probability - mean implied probability)
Mean price (implied probability)
0.1
0.0
-0.1
0.00
0.25
0.50
0.75
1.00

Figure 2: Market prices across probabilities at ten stages of markets' lifecycles. The plots show differences between the market prices and the realized probabilities for 50 equal-sized probability groups. We split each market's trading period into ten consecutive time bins of length 10% each. Within each bin, we compute the market price as the average price of the last 10 trades that occur inside that bin (or fewer if less than 10 trades occur in the bin). Black points represent the default option, points in light gray are the alternative option.

In summary, we find that prices on Polymarket are good estimates of the probabilities of future events most of the time. However, we document a pronounced bias favoring the default option, particularly shortly after the creation of new markets. From a behavioral perspective, this *default bias* is consistent with status quo bias, which describes the reluctance of individuals to deviate from the standard or pre-selected choice (Samuelson & Zeckhauser, 1988). Additionally, the default outcome is often "Yes" and highlighted in green on the website, whereas the complementary outcome "No" is displayed in red, and the price chart is shown only for the probability of the default outcome. This more salient and asymmetric framing likely increases the attractiveness of the default choice, particularly under limited attention (Tversky & Kahneman, 1981; Bordalo et al., 2012; Sims, 2003). Finally, another contributing factor may be cognitive ease, as it is easier to assess the probability that a specific event will occur than the residual probability of *all* other possibilities (Fox and Tversky, 1998). Notably, however, we find no evidence for a market-wide longshot bias.

4.2 Are prediction markets more accurate than bookmakers?

The previous subsection demonstrated that Polymarket prices are, on average, a good estimator of real-world probabilities. However, decision makers who wish to use these predictions for risk management still must consider whether they offer added value over alternative sources of predictions, such as experts and opinion polls. Polymarket emphasizes that its market prices are the most accurate because traders are economically incentivized to process all available information (Polymarket, 2025). Previous research has shown that prediction markets in general (Berg et al., 2008; Wolfers and Zitzewitz, 2004) and Polymarket in particular (Ng et al., 2025) indeed outperform polls. However, another source of predictions, and likely the main competition for bettors, is online bookmakers. In contrast to polling or experts, these bookmakers are also driven by economic incentives when setting the odds for future events.

Thus, this subsection tests whether Polymarket market prices are more accurate than online betting bookmaker odds. We use a subsample of our dataset because we require markets that have identical outcomes on Polymarket and on sports betting websites. Specifically, we focus on tennis matches from the ATP and WTA tours because they are easily identifiable and traceable and have only two possible outcomes (winner and loser of the match). To calculate the implied probabilities of bookmaker odds, we use the mean odds across multiple online betting venues at game start compiled by the British platform *Tennis-Data.co.uk*.[7] Considering the bookmaker's profit margin (i.e., the overround), the implied fair probability p of player A with odds O_A winning the game against player B with odds O_B is:

$$p = \frac{{O_A}^{-1}}{{O_A}^{-1} + {O_B}^{-1}}$$

For example, the match that had the highest betting volume on Polymarket in our sample (USDC 1.8 million), between Jannik Sinner and Ben Shelton in London on July 9, 2025, had mean odds of $O_{Sinner} = 1.29$ and $O_{Shelton} = 3.58$ across sports betting websites at the start of the match. Jannik Sinner thus had an implied probability of winning the game of 73.5%. For Polymarket, we calculate the market price as the rolling average of the last ten trades on the platform (73.6% at game start for the above example).

Then, we match all completed tennis games in the bookmaker dataset with our Polymarket dataset by using the last names of the winners and losers, as well as the date of the match. This yields a total of 492 markets with 984 outcomes until August 2025. Furthermore, to ensure that ex-post information about the course of the game is not included in the analysis, it is crucial to determine the correct start time for the tennis games. For this reason, we manually researched and verified the actual start times for all 492 games instead of relying on the information about the scheduled start time in Polymarket's market descriptions.[8]

[7] The mean odds in the dataset are originally from the website *Oddsportal.com*. We also use the odds from the two individual bookmakers *Bet365* and *Pinnacle Sports* as a robustness test.

[8] The start times indicated by Polymarket were almost always incorrect. Since tennis matches often start late due to the unpredictable length of the previous match on the same court, determining the actual start time is surprisingly

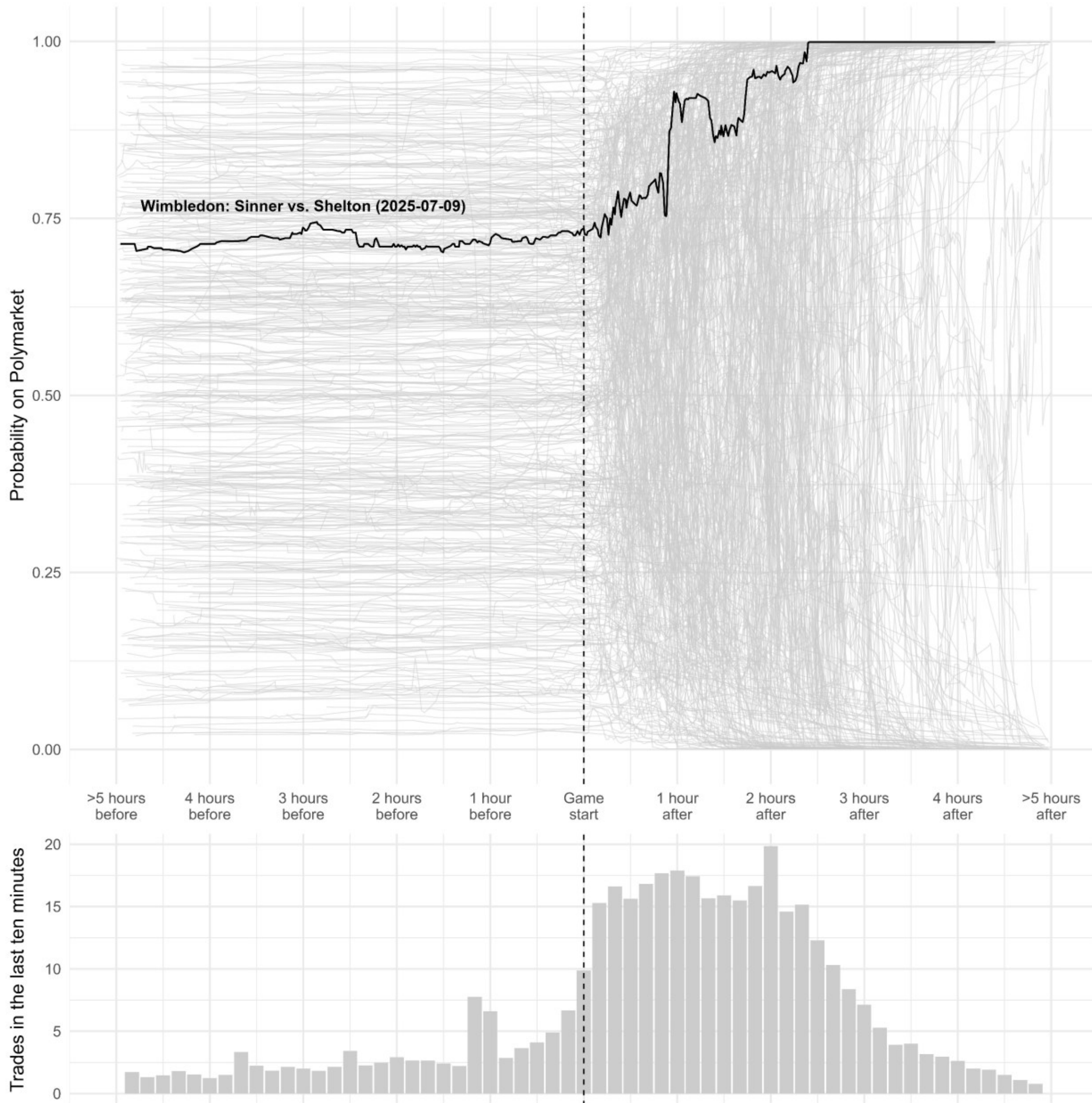

Figure 3: Overview of tennis markets on Polymarket. The upper subplot shows the minute-by-minute development of prices (i.e., probabilities) for the default option on Polymarket for all 492 tennis matches in our sample (in light gray). Before the match begins, the prices remain relatively constant. However, after the match starts, the prices fluctuate more strongly and move towards either 1 (the default option wins) or 0 (the default option loses). As an example, the market for the Sinner vs. Shelton match at Wimbledon on July 9, 2025, is highlighted. The lower subplot shows the average number of trades per 10-minute interval.

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difficult. The different time zones and the fact that long games are sometimes continued after a break on a later date further complicate this process. Therefore, we placed great emphasis on the accuracy of the data and manually collected each value. We primarily relied on the sports statistics website sofascore.com and used additional sources, such as livestreams and live tickers of the games, to verify the start times. Additionally, we rechecked all start times if Polymarket prices at game start deviated by more than 5% from the implied probabilities of the bookmaker odds.

Figure 3 provides a visual overview of these tennis markets. It shows the price development of all 492 matches, from five hours before to five hours after the start of each match (upper subplot; light gray), as well as the average number of trades in ten-minute intervals (lower subplot). The match between Sinner and Shelton discussed above is highlighted. The plot shows that, before the game starts, prices on Polymarket tend to be fairly static because little new information is being processed. Once the game begins, however, the number of trades and price volatility increase significantly. After a few hours, prices approach either 100% (the default option wins; in our example this is Sinner) or 0% (the default option loses). Only a few games remain undecided after five hours, as some are interrupted and continued the next day.

Next, we compare the accuracy of Polymarket predictions with bookmaker odds. We use two simple metrics: a) the mean prediction error and b) the proportion of more accurate predictions. The former is the mean absolute difference between the implied probability and the outcome (0 or 1), while the latter is the percentage of matches in which the Polymarket price was closer to the outcome than the implied probability of the bookmaker odds.

The upper half of Figure 4 shows the results for the two metrics at different points in time before and after game start. The bars at the beginning of the game are highlighted because it is only at this point that the accuracy can be directly compared. The mean prediction error is slightly lower at game start for Polymarket (subplot 1a), and Polymarket's predictions deviate less than the bookmakers' in 69.9% of all games (subplot 1b). Even up to five hours before the match starts, Polymarket slightly outperforms the bookmakers' odds. After the game starts (light grey in the plots), accuracy increases significantly, and the mean error approaches zero, as the winner becomes clearer.

In the lower half of the figure, we further examine the highlighted metrics by grouping the games by probability level at game start to identify potential causes of the differences. Both metrics (subplot 2a and 2b) indicate that Polymarket is much more accurate for bets with extreme probabilities (less than 10% or more than 90%), while the accuracy is nearly identical for moderate probabilities (between 40% and 60%). This is consistent with the observation from the previous subsection that the longshot bias is not pronounced at the market level on Polymarket, though previous research has extensively demonstrated its presence in sports betting (e.g., Ali, 1977; Golec and Tamarkin, 1995; Snowberg and Wolfers, 2010; Franke, 2020).

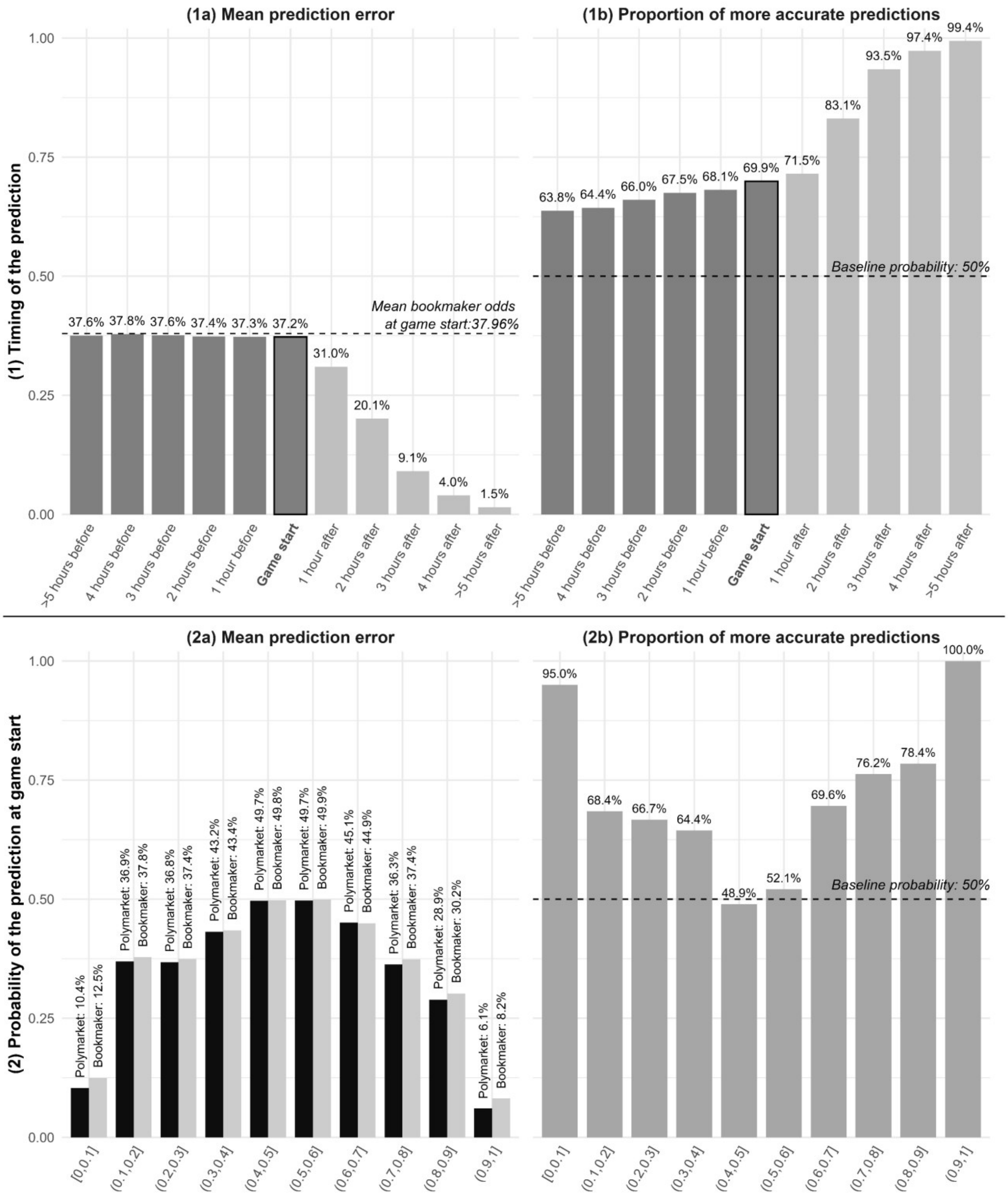

Figure 4: Accuracy of Polymarket predictions compared to bookmakers. Top panel: Polymarket predictions consistently outperform bookmaker odds according to mean prediction error and the proportion of more accurate

predictions. Bottom panel: Polymarket predictions perform substantially better in the tails of the distribution (extreme probabilities), and slightly worse in the center.

As a robustness check, we also separately compared Polymarket prices to the odds from bookmakers *Bet365* and *Pinnacle Sports*. The results are very similar, with Polymarket outperforming these odds for both the mean prediction error (38.01% and 38.36%, respectively, versus 37.2%) and the proportion of more accurate predictions (65.9% and 60%). The longshot bias is less pronounced in the case of *Pinnacle Sports* because it is known as a so-called "sharp bookmaker" (i.e., it has low margins and accepts skilled bettors). However, Polymarket is still more accurate for extreme probability brackets (76.5% and 80% for bets with probabilities lower than 10% and higher than 90%, respectively).

Our finding that Polymarket's accuracy is at least as good as that of bookmakers is notable given that prices in the examined markets are formed by only a small number of trades, and there is likely little insider information available before a tennis match begins. Although the economic benefits of better sporting event predictions are limited, it seems likely that higher liquidity and insider trading make prices even more information-efficient in markets that are important for decision-makers when assessing risks, such as for political or macroeconomic events.

5 Individual level: Profit distribution, skill, and bias

5.1 How are profits and losses distributed across traders?

First, we provide an overview of traders' accumulated profits on Polymarket, focusing on the share of traders who generate profits versus those who incur losses. These proportions are relevant for potential regulation of decentralized prediction markets, as a high share of losing traders may raise concerns regarding the need for consumer protection. It also sheds light on market efficiency: a high prevalence of unprofitable traders can indicate substantial noise trading, which may distort prices and reduce the platform's predictive accuracy.

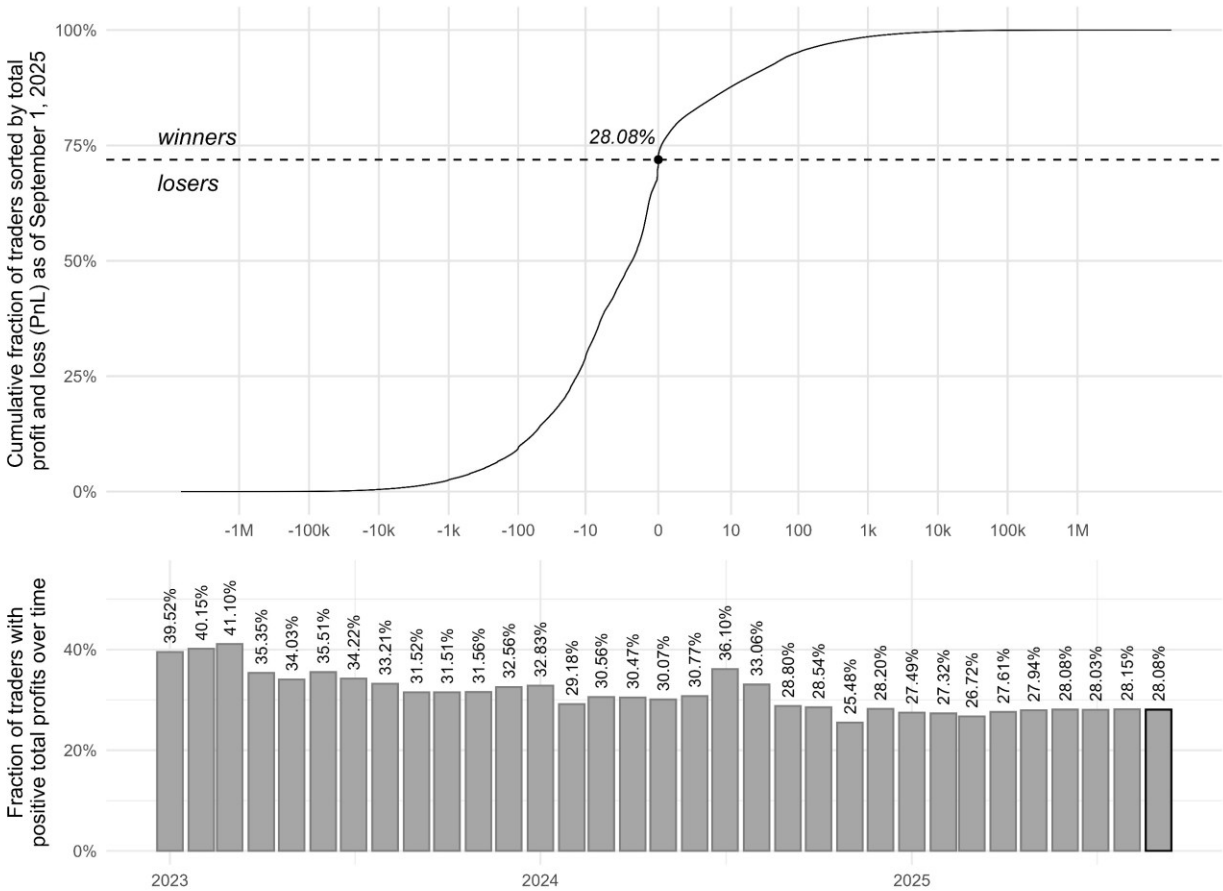

Figure 5: Distribution of profits and losses, and fraction of winning traders. The top panel shows the shares of winners and losers according to total profits at a logarithmic scale. The bottom panel shows the fraction of traders with positive total profit across months. The last column is highlighted because it corresponds to the upper subplot.

The upper panel in Figure 5 depicts the distribution of accumulated profits and losses among traders as of September 1, 2025. We use a logarithmic transformation for both profits and losses, as trader profits are up to USDC 22M and losses go up to absolute values of USDC 6M. We find that less than 30% of traders earn positive profits. Most traders feature moderate profits and losses between + USD 100 and – USD 100. However, there are also several traders with substantial profits or losses of above USDC 1,000, as 1% already represents roughly 10,000 traders.

Next, we examine whether the distribution of winners and losers has changed over time. The lower panel shows the fraction of traders with positive total profits and losses until the respective month from January 2023 to September 2025. It is evident that the fraction of winning traders is consistently below

50%. Furthermore, the fraction of winning traders decreases over time, indicating that the rapid growth of Polymarket (see Figure 1 in Section 3) has led to an increase in the proportion of less skilled traders.

5.2 Are there skilled traders?

Section 5.1 indicates that more traders experience losses than achieve profits. Next, we explore whether this distribution is the result of randomness, i.e., luck, or whether there is evidence that traders have different levels of skill.

Our test of non-random skill proceeds as follows. We examine the gain per bet of each trader, i.e., the difference of the buying price and outcome for all settled markets at the reference date (September 1, 2025). Skilled traders should realize a positive mean gain, which we test using a t-test. However, as traders often trade multiple times in the same market, the independence assumption of the t-test may be violated. To address this, we run t-tests for each trader on all trades while clustering standard errors at the market level, allowing for dependent observations within a market.[9]

Under pure randomness, we would expect the p-values of our t-tests to be uniformly distributed. For instance, we would expect roughly 1% of traders to have achieved gains that only have a 1% chance to show up under a distribution with a mean gain of zero. By comparing our empirical cumulative distribution function (CDF) with the theoretical uniform distribution, we can see whether skill beyond luck is present in the data. Specifically, if there are traders with a positive expected gain per bet, we would expect the empirical distribution of p-values to be above the theoretical CDF for low p-values. Furthermore, there may be traders with negative skill, for instance due to biases. Therefore, we expect empirical p-values below the theoretical ones in the right tail of the distribution, as this implies a higher share of traders with negative mean gain per bet. As t-tests require at least a couple of observations, we focus our analysis on regular

[9] As robustness check, we instead aggregate all trades of a trader within a market by computing the contract-weighted mean of the gain variable. This produces a dataset with one observation per market and trader, for which independence is more plausible. Results are qualitatively similar, suggesting that our findings are not driven by autocorrelation from multiple trades within the same market.

traders, i.e., those with bets on at least 25 different markets (n = 180,101 traders). Figure 6 presents the results with standard errors clustered at the market level.

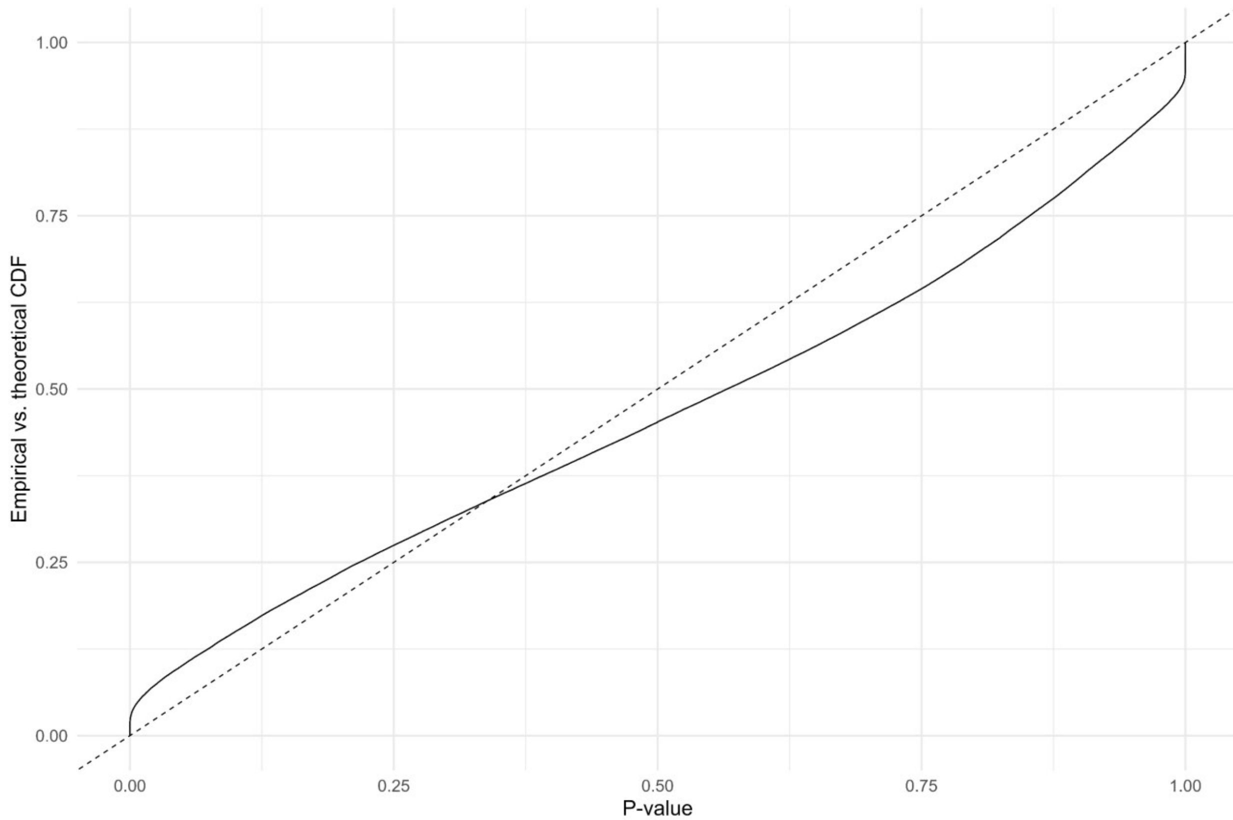

Figure 6: Empirical vs. theoretical CDF for individual trades and clustered standard errors at the market level. The dashed line shows the theoretical CDF of p-values. The empirical p-values (solid line) are above it for low p-values, and below it for high p-values. This indicates that traders with significant skill (and negative skill) exist.

The observed pattern is in line with our prediction that there is skill beyond luck. In the left tail, the empirical p-values are roughly 3% above the theoretical CDF, indicating that 3% of regular traders have achieved gains per bet that are unlikely to be observable under a distribution without positive skill. In the right tail, the empirical CDF is roughly 5% below the theoretical one, indicating negative skill of a substantial fraction of traders. Therefore, our t-tests using the gain per bet indicate that a substantial fraction of traders display skill beyond luck. To further substantiate this idea, we investigate the persistence of profits and losses and winning probabilities in the next section.

5.3 *Are profits and losses persistent over time?*

To test whether profits and losses are persistent, we examine the relationship between a trader's monthly PnL and their PnL in the previous month. For traders who do not trade every month, we replace the previous month's PnL with the most recent month in which they had trading activity.

We first conduct a one-sided Welch two-sample t-test comparing traders in the top and bottom PnL decile of the previous month. To ensure the results are not driven solely by differences in trading activity, we restrict the sample to traders in the top quartile of trade frequency. The results of the Welch t-test indicate that traders in the top PnL decile of the previous month achieved significantly higher monthly profits (mean = 906.10) than traders in the bottom decile (mean = -691.67). The t-statistic is 5.14 with 70,644 degrees of freedom, indicating statistical significance at the 0.1% level. The lower bound of the one-sided 95% confidence interval is USDC 1086.94, implying that, on average, traders with previous gains earn at least USDC 1086 more than traders with previous losses. These results provide evidence that profits and losses are persistent across months.

As a second test, we use logistic regression to estimate the probability that a trader achieves a positive PnL in a given month, conditional on having achieved a positive PnL in their previous month of trading. To account for the fact that bets with higher implied probabilities are more likely to win, we include the trader's contract-weighted mean bet probability as a control variable.

	Estimate	**SE**	**z-value**
Intercept	-1.3393	0.0061	-218.9***
Previous winner	**0.8041**	**0.0031**	**256.9*** **
Mean implied probability	1.8328	0.0120	152.5***

Table 2: Logistic regression of winning probability per month. Logit regression of winning frequencies on a dummy (positive profit in the previous month with trading activity) and the mean played probability as control.

In the logistic regression of winning probability in Table 2, we find that previous winners have a significantly higher chance of achieving positive PnLs in the next month. For example, at a traded mean implied probability of 50%, the estimated probability to earn a positive PnL for a trader with previous gains is $\frac{1}{1+\exp\left(1.3393-0.8041\cdot 1-1.8328\cdot 0.5\right)}=59.42\%$, while it is $\frac{1}{1+\exp\left(1.3393-0.8041\cdot 0-1.8328\cdot 0.5\right)}=39.58\%$ for a trader with previous losses. This means that a positive PnL in the previous month increases the chances of winning by roughly 19.84 percentage points. Therefore, the PnLs of traders are persistent in terms of both absolute PnL and winning probabilities, providing further support for the existence of trader skill.

5.4 What distinguishes winners and losers?

Finally, our dataset allows us to examine the characteristics of winners and losers on Polymarket. Differences in these characteristics may help explain variations in trading performance and provide a foundation for research on profitable strategies in decentralized prediction markets, as well as models for identifying skilled traders. Furthermore, this analysis enables us to examine whether the trading behavior of top traders vs. bottom traders is associated with the biases identified in Section 4.

We divide traders into top and bottom quantiles based on total profits and losses as of September 1, 2025 (consistent with Figure 5 in Section 5.1) and compare the characteristics for both groups. Table 3 provides the results for three alternative definitions of top and bottom traders (10%, 1%, and 0.1% quantiles). Furthermore, Table 4 offers an analysis based on a subset of traders with high trading activity according to the number of unique markets, total number of contracts, and total amount of USDC traded.

While t-statistics are reported in both tables to indicate statistical significance, they might be inflated by the large sample size of up to 100,000 traders per group. Therefore, Cohen's *d* is also reported as a measure of effect size (Cohen, 2013). Cohen's d is the difference in means divided by the pooled standard deviation in both groups. This means that it is an indication of whether the observed differences in means, i.e., variations between groups, are substantial relative to the variations within groups. For example, in Table 3, Cohen's d amounts to more than 0.50 for the contract-weighted proportion of bets on favorites ($p>0.50$), indicating that the difference of 13 percentage points is not only statistically significant but also substantial relative to the variance within groups, indicating that there is limited overlap between the distributions in both groups. Conversely, while the difference in the number of trades (*# trades*) is economically relevant (675 vs. 243) and statistically significant, the corresponding Cohen's d is small. This reflects considerable variance within the top and bottom deciles, indicating that there is a major overlap of the distributions of the number of trades between these two groups. The key takeaways from the two tables are as follows.

Probabilities. Traders in the top PnL decile have a higher fraction of bets on favorites (*p>0.50, p>0.75, p>0.90, p>0.95, p>0.99*), a higher mean implied probability (*Mean(*p*)*), greater variance in implied probabilities (*Var(p)*), and a larger share of extreme longshots (*p<0.01*). The higher proportions of bets on favorites across all probability ranges are associated with very high *t*-values and substantial effect sizes (Cohen's $d > 0.3$). This indicates that well-performing traders play more favorites than traders who lose money, potentially exploiting the favorite-longshot bias. The higher mean implied probability reinforces this interpretation. The greater variance in implied probabilities among top traders suggests that they do not specialize in a narrow range of bets but rather participate across the probability spectrum, likely focusing on wagers with favorable prices. Both effects are statistically significant and exhibit meaningful effect sizes (d > 0.3). Interestingly, the fraction of extreme longshots with implied probabilities below 1% is also higher in the top decile (Cohen's d of 0.16). This may reflect skilled traders' ability to identify and act on rare opportunities, for example, by accessing information early or firsthand (e.g., attending relevant events).

The results are similar when defining top and bottom traders based on percentiles or 0.1%-quantiles instead. There are two differences. First, based on percentiles, the fractions of favorites between 0.90 and 0.95 are not higher. Second, based on 0.1%-quantiles, all fractions of probability intervals are positive and significant, indicating that traders from the bottom decile trade more bets in the 25% to 50% interval of implied probabilities. The directions and values of Cohen's d for the main differences, i.e., more favorites (*p>0.50*), more extreme longshots (*p<0.01*), and higher variance of implied probabilities (*Var(p)*), are consistent and substantial across all specifications of top and bottom traders. Furthermore, in Table 5, the results are similar but the differences regarding extreme longshots are less substantial.

Variable	Deciles, n = 97,364 Top Mean	Bottom Mean	Cohen's d	t-value	Percentiles, n = 9,736 Top Mean	Bottom Mean	Cohen's d	t-value	0.1%-Quantiles, n = 973 Top Mean	Bottom Mean	Cohen's d	t-value
Total PnL as of September 1, 2025	3823.31	-3427.32			36257.29	-29210	-	-	285300.86	-186524	-	-
Mean implied probability (p)	0.4695	0.3849	0.47**	103.47***	0.4488	0.4317	0.1	6.94***	0.4860	0.4815	0.03	0.6
Variance of p	0.1103	0.0785	0.4**	89.06***	0.0839	0.0771	0.11*	7.94***	0.0871	0.0723	0.29*	6.29***
Fraction of p > 0.99	0.1286	0.0722	0.33**	71.78***	0.0561	0.06	-0.03	-2.27*	0.0846	0.0629	0.17*	3.67***
Fraction of p > 0.95	0.1928	0.1151	0.39**	85.14***	0.1083	0.1116	-0.02	-1.43	0.1492	0.1181	0.17*	3.83***
Fraction of p > 0.90	0.2232	0.1399	0.39**	86.51***	0.1439	0.1445	-0.0006	-0.22	0.1865	0.1571	0.15*	3.26**
Fraction of p > 0.75	0.2831	0.1973	0.37**	82.33***	0.2242	0.217	0.03	2.3*	0.2630	0.2325	0.14*	2.98**
Fraction of p > 0.50	0.4864	0.3568	0.52***	114.82***	0.4662	0.4086	0.22*	15.69***	0.4978	0.4639	0.14*	3.02**
Fraction of p < 0.25	0.3520	0.4433	-0.31**	-68.44***	0.3384	0.3612	-0.08	-5.87***	0.3039	0.2683	0.15*	3.31**
Fraction of p < 0.10	0.2901	0.3278	-0.13*	-28.91***	0.2468	0.2463	0.0005	0.16	0.2038	0.1775	0.13*	2.91**
Fraction of p < 0.05	0.2531	0.2591	-0.02	-4.73***	0.1992	0.1821	0.08	5.3***	0.1629	0.1295	0.19*	4.19***
Fraction of p < 0.01	0.1729	0.1364	0.16*	34.68***	0.1126	0.0837	0.18*	12.32***	0.0963	0.0614	0.28*	6.14***
Mean trade percentile (t)	0.7203	0.7493	-0.15*	-32.36***	0.7482	0.77	-0.12*	-8.38***	0.7834	0.8092	-0.17*	-3.71***
Fraction of t > 0.90	0.3780	0.4436	-0.18*	-39.89***	0.4458	0.4903	-0.13*	-9.33***	0.5052	0.5495	-0.14*	-3.15**
Fraction of t > 0.75	0.5780	0.6321	-0.16*	-34.59***	0.631	0.6649	-0.11*	-7.78***	0.6918	0.7297	-0.14*	-3.18**
Fraction of t > 0.50	0.7712	0.8109	-0.14*	-31.85***	0.8066	0.831	-0.11*	-7.53***	0.8462	0.8759	-0.16*	-3.6***
Fraction of t < 0.25	0.0821	0.0881	-0.03	-7.55***	0.0907	0.0813	0.06	4.24***	0.0724	0.0591	0.11*	2.5*
Fraction of t < 0.10	0.0311	0.0399	-0.08	-17.68***	0.0396	0.0343	0.06	3.94***	0.0312	0.0257	0.08	1.75
Fraction on yes	0.5317	0.5962	-0.25*	-55.19***	0.5432	0.5899	-0.2*	-13.62***	0.5335	0.5784	-0.2*	-4.41***
Fraction on default	0.5437	0.6059	-0.27*	-60.05***	0.5533	0.5970	-0.2*	-14.28***	0.5474	0.5846	-0.19*	-4.1***
Number of trades	675.09	242.82	0.04	9.01***	5611.57	1286.39	0.13*	9.18***	28305	4830.97	0.28*	6.28***
Number of markets	42.3701	35.6051	0.03	7.39***	174.258	93.0651	0.14*	10.02***	565.47	190.9579	0.32**	6.95***
Number of contracts	237789.7	131147.1	0.03	7.66***	2018833	857339.5	0.12*	8.54***	12991558	4250871	0.33**	7.29***
Total USDC traded	86511.89	43636.27	0.04	8.39***	773231	327066.3	0.13*	8.91***	5496514	1840824	0.36**	8.02***
USDC per bet	358.03	349.7119	0	0.86	1950.04	1678.99	0.04	3.07**	5251.60	4282.99	0.07	1.48

Table 3: Characteristics of traders from top and bottom quantiles of PnL. Means, variance and fractions are calculated using weights based on the number of contracts. We define top and bottom traders as traders from the top and bottom decile, percentile, and 0.1%-quantile, respectively. The specified n is the number of traders per group. Top mean and bottom mean represent the contract-weighted means of each group of traders. Statistical significance is indicated by t-values and asterisks (p-values of *** <0.001, ** < 0.01, * < 0.05), while Cohen's d is an indication of effect size, i.e., whether effects are substantial relative to in-group variation (d of *** > 0.5, ** > 0.3, * > 0.1). The mean profits and losses (PnL) of the groups are included as a reference.

	Number of Markets, n = 9,837				Number of Contracts, n = 9,736				Total USDC Traded, n = 9,736			
Variable	**Top Mean**	**Bottom Mean**	**Cohen's d**	**t-value**	**Top Mean**	**Bottom Mean**	**Cohen's d**	**t-value**	**Top Mean**	**Bottom Mean**	**Cohen's d**	**t-value**
Total PnL as of September 1, 2025	16947.2	-11570.6	-	-	35531.9	-28697.9	-	-	35737.2	-28472.9	-	-
Mean implied probability (p)	0.4761	0.4389	0.26*	18.49***	0.4631	0.4167	0.27*	18.97***	0.4763	0.4572	0.12*	8.55***
Variance of p	0.1158	0.1053	0.17*	11.74***	0.1043	0.0877	0.27*	18.77***	0.0962	0.0893	0.12*	8.29***
Fraction of p > 0.99	0.1171	0.0728	0.3*	20.86***	0.0815	0.0665	0.11*	7.85***	0.0657	0.0607	0.04	2.94**
Fraction of p > 0.95	0.1841	0.1318	0.3*	20.95***	0.1529	0.1235	0.17*	11.79***	0.1351	0.1196	0.09	6.43***
Fraction of p > 0.90	0.2224	0.1701	0.28*	19.81***	0.1964	0.1588	0.2*	13.76***	0.1811	0.1595	0.11*	7.98***
Fraction of p > 0.75	0.2974	0.2536	0.22*	15.78***	0.2802	0.23	0.24*	16.72***	0.2759	0.2462	0.14*	9.56***
Fraction of p > 0.50	0.4835	0.4407	0.24*	17.04***	0.4737	0.3957	0.35**	24.49***	0.4955	0.4480	0.22*	15.24***
Fraction of p < 0.25	0.3480	0.3844	-0.16*	-11.47***	0.3610	0.4092	-0.18*	-12.86***	0.3290	0.3370	-0.03	-2.39*
Fraction of p < 0.10	0.2712	0.2885	-0.08	-5.64***	0.2735	0.2963	-0.09	-6.27***	0.2431	0.2309	0.06	4***
Fraction of p < 0.05	0.2308	0.2354	-0.02	-1.57	0.2240	0.225	0	-0.28	0.1967	0.1759	0.11*	7.39***
Fraction of p < 0.01	0.1516	0.1400	0.06	4.5***	0.1315	0.1041	0.16*	10.81***	0.1116	0.0883	0.16*	10.84***
Mean trade percentile (t)	0.7051	0.7261	-0.14*	-9.8***	0.7342	0.762	-0.16*	-11.14***	0.7417	0.7545	-0.08	-5.25***
Fraction of t > 0.90	0.3418	0.3878	-0.19*	-13.34***	0.4199	0.4755	-0.18*	-12.37***	0.4350	0.4583	-0.08	-5.27***
Fraction of t > 0.75	0.5713	0.6049	-0.13*	-9.1***	0.6102	0.6508	-0.14*	-9.77***	0.6210	0.6420	-0.07	-5.16***
Fraction of t > 0.50	0.7683	0.7915	-0.11*	-8.04***	0.7939	0.8233	-0.13*	-9.4***	0.8010	0.8163	-0.07	-5.05***
Fraction of t < 0.25	0.1051	0.0996	0.04	2.92**	0.0990	0.0851	0.09	6.46***	0.0960	0.0880	0.06	3.91***
Fraction of t < 0.10	0.0460	0.0456	0.01	0.41	0.0425	0.0366	0.07	4.55***	0.0412	0.0371	0.05	3.41***
Fraction on yes	0.5061	0.5504	-0.22*	-15.32***	0.5357	0.5849	-0.24*	-16.7***	0.5295	0.5644	-0.18*	-12.31***
Fraction on default	0.5248	0.5542	-0.23*	-16.15***	0.5375	0.5869	-0.27*	-18.76***	0.5317	0.5669	-0.21*	-14.46***
Number of trades	6042	1765.78	0.13*	9.11***	5962	1414.00	0.14*	9.6***	5868	1500.79	0.13*	9.21***
Number of markets	247.70	190.51	0.1	6.79***	202.68	106.166	0.17*	11.74***	194.03	118.30	0.13*	9.15***
Number of contracts	1912653	728207	0.13*	8.85***	2102695	897109	0.13*	8.85***	2076942	889066	0.12*	8.72***
Total USDC traded	710044	291990	0.12*	8.58***	795636	336686	0.13*	9.15***	791389	344300	0.13*	8.92***
USDC per bet	338.98	320.44	0.02	1.07	1753.32	1519.5	0.04	2.63**	1945.61	1529.30	0.07	4.67***

Table 4: Characteristics of traders from top and bottom deciles of PnL, filtered by trading volumes. We define top and bottom traders as traders from the top and bottom decile, percentile, and 0.1%-quantile, respectively. In this table, we limit the analysis to traders from the top deciles according to the number of traded markets, number of contracts, and total USDC amount traded, respectively. The specified n is the number of traders per group. Top mean and bottom mean represent the contract-weighted means of each group of traders. Statistical significance is indicated by t-values and asterisks (p-values of *** <0.001, ** < 0.01, * < 0.05), while Cohen's d is an indication of effect size, i.e., whether effects are substantial relative to in-group variation (d of *** > 0.5, ** > 0.3, * > 0.1). The mean profits and losses (PnL) of the groups are included as a reference.

Timing. Top-decile traders tend to trade earlier in a market's lifecycle: their average trade-time quantile is lower, and they trade less frequently in the second half of markets ($t>0.50$, $t>0.75$, $t>0.90$). These differences are statistically significant but only show low to moderate effect sizes, indicating that timing systematically differs across groups but with substantial within-group variation. Under stricter definitions of top traders, they also trade more in the earliest 10% and 25% of market duration ($t<0.10$, $t<0.25$). This pattern is consistent with skilled traders strategically exploiting early-stage mispricing (see Section 4.1), thereby moving market prices toward true probabilities and contributing to overall market efficiency. Surprisingly, top traders trade less in the final 10% of market duration, despite the systematic pricing patterns documented in Section 4.1. This suggests that deviations from true probabilities in later market stages might persist because of an increased presence of unskilled traders.

Bets on the "Yes" and the default option. The mean proportion of bets on "Yes" and the default option are more than 6 percentage points lower for traders from the top decile, which is both statistically and economically significant. The effect size is moderate ($d = 0.25$ and 0.27), suggesting that better-performing traders are less susceptible to overinvesting in the default choice.

Similar differences can be seen in the other definitions of top and bottom traders, and Table 4. This indicates that both the default bias and the tendency to overtrade the "Yes" option contribute to profits and losses, and therefore are a potential source of inefficiencies in decentralized prediction markets.

Other characteristics. As expected, the number of trades, number of unique markets, and total number of contracts, and USDC traded are higher among top-decile traders. However, Cohen's d values indicate that within-group variation is more substantial than differences between groups for these variables. Surprisingly, the average USDC bet size is almost identical across groups ($t < 1$), with consistently low effect sizes across all specifications. For the other variables, effect sizes become larger when stricter definitions of top and bottom traders are used, with Cohen's d exceeding 0.3 in the 0.1% quantile comparisons. A similar pattern appears when focusing on subsets of high-activity traders. These results suggest that experience and diversification may contribute to profit differences and signal underlying skill.

The previous analysis has shown that the portfolios of top traders according to total PnL play a higher fraction of favorites and heavy longshots, trade a wider range of probabilities, tend to trade earlier, bet less on "Yes" and the default option, and trade more than bottom traders. While this suggests that top traders exploit biases of bottom traders, the analysis of differences between top and bottom traders does not account for whether the choice of characteristics actually contributes to the higher profitability of top traders. To address this issue, we also created two subsets of our trade data by filtering for all the trades of top and bottom traders, respectively. We then ran the same analyses as in Section 4.1, to see whether trades with certain characteristics regarding probabilities, timing, and default option, i.e., potential biases, contribute to profits. Figure 7 can thus be interpreted similarly to Figure 2 in Section 4.1, except here we distinguish between trades made by the bottom 10% of traders (left column) and the top 10% of traders (right column).

First, for early trades (t = 10%), we see that buying the default option is associated with lower mean differences between realized and implied probabilities than trading the non-default option for both top and bottom traders. This shows that the lower fraction of bets on the default option of top traders is likely to positively contribute to their higher profits, indicating that these may exploit the bias (below-average performance) of bottom traders. The highest positive differences can be observed for extreme longshots on the non-default option for top traders, suggesting that their high fraction of extreme longshots likely is not a bias but a means of taking advantage of mispricing. Over the lifecycle of a market, the differences between the default option and non-default option diminish, in line with the notion that skilled traders exploit the bias, and, in this way, move the market towards efficiency (see also Section 4.1).

Regarding probabilities, extreme longshots (small probabilities) perform relatively well across all timing stages, indicating that the higher fraction of these in the portfolios of top traders is a contributor towards their profits. In the later stages of markets, moderate longshots perform worse than favorites.

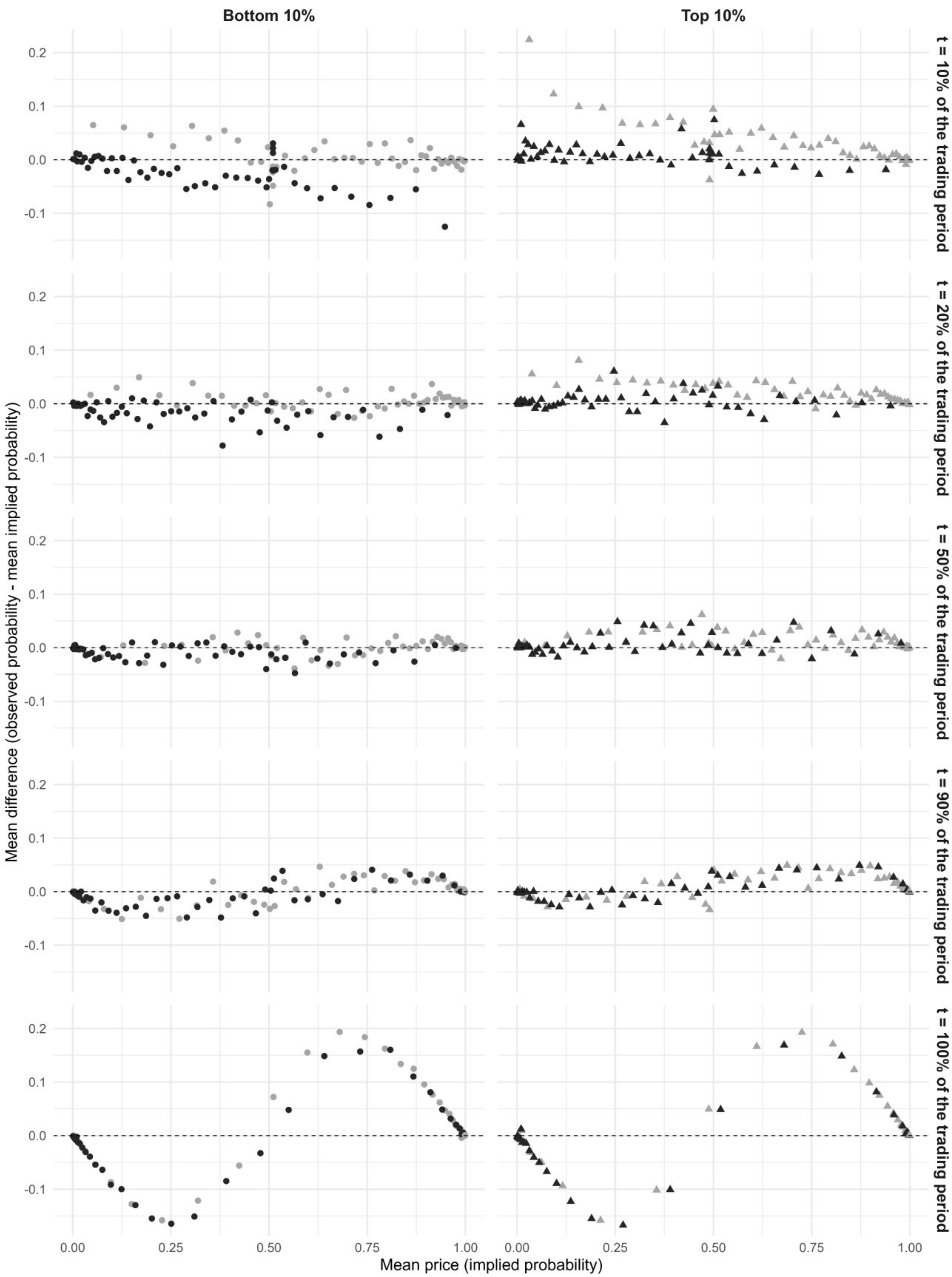
Bottom 10%
Top 10%
t = 10% of the trading period
t = 20% of the trading period
t = 50% of the trading period
t = 90% of the trading period
t = 100% of the trading period
Mean difference (observed probability - mean implied probability)
Mean price (implied probability)
0.2
0.1
0.0
-0.1
0.00
0.25
0.50
0.75
1.00

Figure 7: Prediction accuracy of trades by traders in the top and bottom PnL deciles. As Figure 2, the plots show differences between the market prices and realized probabilities. Black points represent the default option, points in light gray are the other option. On the left the trades from the bottom trader decile according to total PnL is depicted (points), while the right represents the trades from top traders (triangles). Early trades of top traders on the non-default option are highly profitable. In later stages, favorites contribute to the positive PnL. As top traders make more of these trades, we conclude that they likely take advantage of biases of bottom traders.

Notably, for t = 90% of the trading period, the distribution for top traders closely resembles the distribution of bottom traders. Specifically, moderate longshots of top traders are also associated with losses, while favorites are also associated with profits for bottom traders. This pattern is magnified in the last trades of a market. The default option still performs worse, but we observe a strong S-shape that looks similar for both trades of top and bottom traders. Thus, favorites appear to be heavily undervalued, while longshots are heavily overvalued. As previously established, top traders play a higher fraction of favorites, which positively contributes to their profits. A likely explanation of the strong S-shaped pattern is that new information on the event outcome is available (or the event outcome is already determined) but only to some of the traders. Consequently, an event with a current implied probability of, for instance, 75% is already certain, meaning that the correct price is USDC 1 and USDC 0 for the opposite outcome. Skilled traders can use this information by being the first to make trades based on it, exploiting that other traders have not obtained the information yet, or have not acted and canceled their limit orders.

Overall, the analysis suggests that top traders exploit biases and the lower speed of bottom traders to generate profits. This indicates that the observed differences are a promising starting point for future research to identify profitable strategies, skilled traders, and insiders.

6 Discussion

Due to financial incentives of market participants, prediction markets promise to provide highly accurate estimates of event probabilities (Arrow et al. 2008). Polymarket (2025) similarly states that their markets "reflect *accurate, unbiased, and real-time* probabilities". While we document several biases, including a persistent tendency to overtrade the default or "Yes" option, prices are usually good estimators of true real-world probabilities. Biases in the probability dimension also exist, although they are weaker and concentrated in particular stages of a market's lifecycle. Thus, we do not find evidence of a market-wide longshot bias that has consistently been found in traditional sports betting (e.g., Snowberg and Wolfers, 2010). In contrast, extreme longshots, i.e., bets on extremely small probabilities, appear to perform consistently well. One potential explanation is that traders systematically underestimate the likelihood of rare events, consistent with evidence that individuals struggle to assess tail risks (Taleb, 2007). Furthermore, arbitrageurs face substantial risk when attempting to correct mispricing of extreme longshots, which may allow such deviations to persist.

We also find that skill and thus profits and losses vary significantly between traders. In line with the idea that markets require only a sufficient number of skilled traders to move them towards the efficient frontier (Grossman and Stiglitz, 1980; Forsythe et al., 1992), prices on Polymarket are generally accurate and also slightly better than bookmaker odds in tennis matches. Market accuracy is lower at creation and near the resolution of markets, consistent with the notion that skilled traders gradually incorporate available information into prices. This process is likely slowed by the presence of less skilled traders and the inherent inertia of the limit order book, as orders need to be actively changed or canceled.

6.1 Implications

Managerial implications. Decentralized prediction markets offer estimates that are generally accurate and outperform other common information sources. However, managers should be careful when using real-time estimates at the beginning or end of the trading period as processing and incorporating new information takes some time. Furthermore, the presence of biases indicates that estimates can be further improved by adjusting for these (in particular the default bias).

Implications for regulators. We find that fewer than 30% of traders earn positive net profits and that the distribution of profits and losses is highly skewed, which indicates that a large number of traders systematically lose money to a small minority of skilled participants. This pattern emphasizes the importance of consumer protection, which may be achieved by adapting established procedures from gambling regulation, such as warning notices on the platform that a high fraction of market participants loses money. It also appears likely that, in the future, many casual gamblers will (partially) migrate to prediction markets due to their advantages over traditional online betting (e.g., greater variety, lower trading costs, and a secondary market; see Table 1 in Section 2.1). This might lead to an even higher share of unskilled gamblers, reinforcing the relevance of implementing measures of consumer protection.

Implications for market participants. We show that several biases exist. Awareness of these biases may allow market participants to mitigate potential losses. Moreover, our result that predictions are highly accurate demonstrates that market participants should not expect to earn positive profits without specific expertise or informational advantages. Traders lacking such advantages should therefore only trade for the non-monetary utility derived from betting itself, rather than from an expectation of earning profits.

Academic implications. Our study introduces decentralized prediction markets as a novel data source for research in behavioral economics and financial markets. Due to the transparency of the blockchain, the data are of high granularity and allow researchers to study trading outcomes at the individual level, which is usually impossible in financial markets. Our dataset includes trades from nearly one million investors, far more than any laboratory experiment can realistically include. Thus, decentralized prediction markets enable researchers to empirically test hypotheses derived from theoretical models or experiments in a real-world setting with high stakes.[1] Furthermore, the dataset can be used to understand how individuals contribute to the functioning of financial markets. Its high granularity allows multidimensional analyses, such as our analysis of prediction errors across probability and timing ranges, default vs. non-default option, and top vs. bottom traders, which can potentially challenge findings derived from less granular data.

[1] Note that, in contrast to experiments, we cannot control for all trader characteristics. For instance, their total wealth, age, gender, and motives, such as hedging, are unclear.

6.2 Future Research Avenues

Our study is of an explorative nature, meaning that several open questions remain. Identifying and emphasizing these is one of the contributions of this article, as it is intended as a foundation for future research addressing these questions.

For example, we have focused on trader characteristics in our analysis and linked them to biases. Future research could link accuracy and bias to market characteristics, such as the number and volume of trades, and the number of traders, to investigate whether larger markets are more accurate (more skilled traders move the market closer to efficiency), or less accurate due to more unskilled traders who introduce noise and bias. Another direction is studying (behavioral) anomalies, such as momentum, salience, min/max effects, or recency bias, and the interaction between market participants, i.e., how traders react to, for example, large trades of others. Relatedly, researchers can use the data to study how traders learn from their own performance history or the historical performance of peers. Another relevant question is why there is trading on decentralized prediction markets in the first place, as unskilled and uninformed traders can expect to suffer losses by only trading with better informed and skilled traders (no-trade theorem; Milgrom and Stokey, 1982). A possible avenue in this regard is identifying different groups of traders and studying their preferences towards risk. Relatedly, research could identify insiders and investigate their influence on market accuracy to test the strong efficient market hypothesis (Fama, 1970).

Another promising direction is to study price patterns and ways to exploit them to generate profits. Relatedly, future research could study whether it is profitable to follow other traders, and how to identify the set of traders to follow. In this context, new business models that are currently emerging based on decentralized prediction markets, such as *Polyfunds*, which enables investing in the portfolios of traders on Polymarket, can be assessed.

Furthermore, data from decentralized prediction markets offers several opportunities for combining it with other timely datasets. These include information sources, such as news and social media; other betting providers, for investigating arbitrage possibilities; and corporate and household portfolios, for assessing the utility of prediction markets as a means of insurance, i.e., hedging political and macroeconomic risks that are not easily hedged with standard financial instruments.

Finally, decentralized prediction markets offer research opportunities beyond economics and management. For example, in social science, the data could be used to assess whether an event, such as a public speech by a politician, is perceived as positive or negative, and how it affects probabilities of other events such as an election. Relatedly, researchers could study how political views affect expectations. For example, Eichengreen et al. (2025) use data from Polymarket to study how beliefs regarding central bank independence affect monetary policy. Furthermore, the implied probabilities of events can be used as an input in social science research, for instance as a representation of the population's optimism or pessimism regarding economic development or geopolitical risks.

7 Conclusion

Decentralized prediction markets offer granular data at the individual level that can add an important new dimension to behavioral research. In this study, we explore decentralized prediction markets based on a dataset of more than 124 million trades on Polymarket. We demonstrate that Polymarket's predictions are generally accurate and slightly outperform the odds of bookmakers. At the same time, we document several biases, such as a tendency to overtrade the default and "Yes" option, and patterns of mispricing that likely reflect the time needed for markets to incorporate new information. At the individual level, we show how profits and losses are distributed across traders, provide evidence on skill beyond luck, and identify systematic differences between top and bottom traders according to total profits. Our results suggest that skilled traders are able to exploit biases and the lower speed of unskilled traders. Finally, we discuss implications and outline avenues for future research, enabling researchers to leverage this largely untapped source of data for research in behavioral economics, finance, market design, and other disciplines.

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